# Can LLMs Read Polls?

## Automating the Extraction of Transparency Metrics from Text Descriptions of Polling Methods

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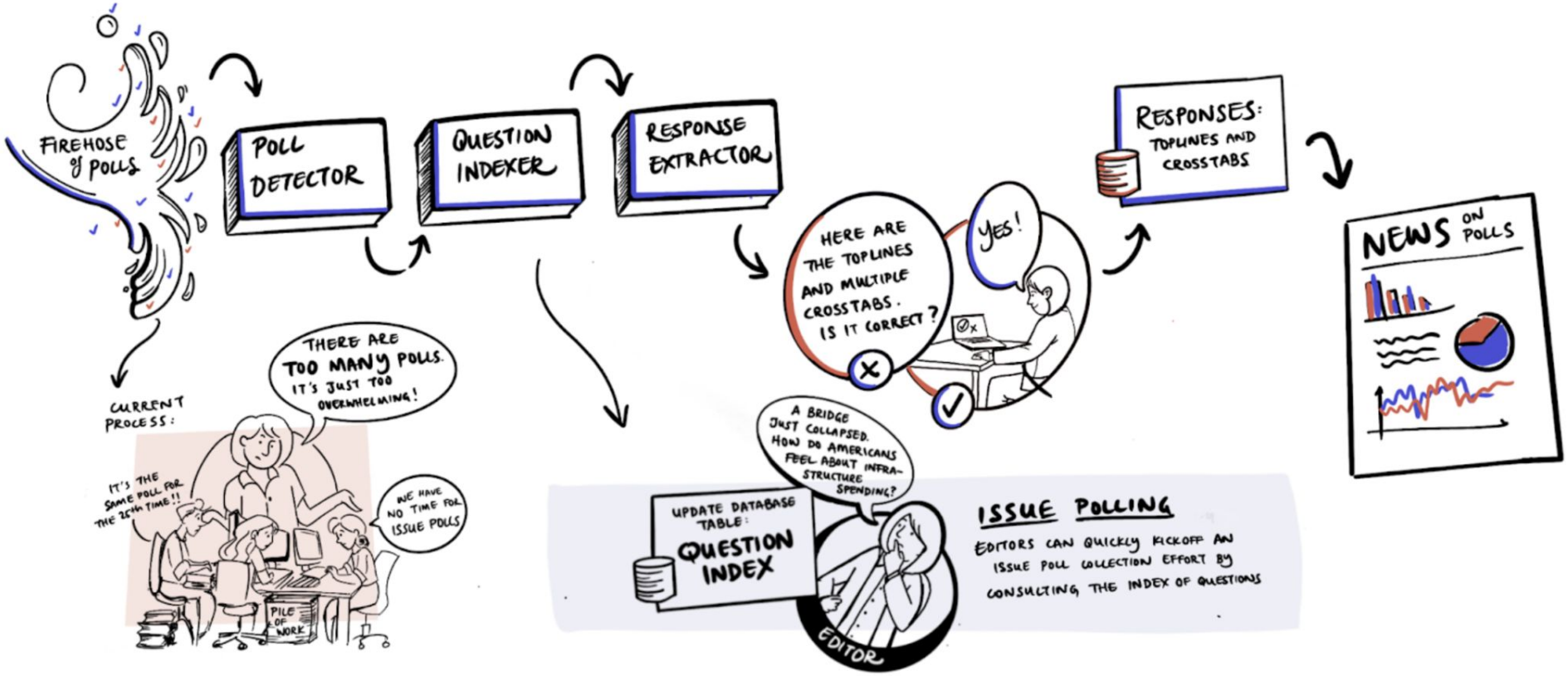
# A small part of our pollfinder.ai project

Use LLMs to find and index all public opinion polls for a **polling unit** of a newsroom.



# Transparency Scores

**Measuring Poll Transparency is Important**

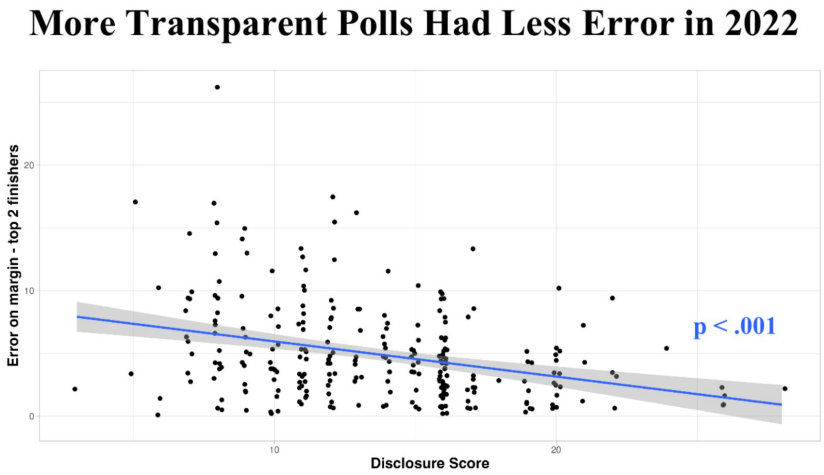
Poll transparency builds public trust and promotes media accountability in the use of survey data.

**Challenges**

Polling methodologies are often reported in unstructured formats

Manual extraction is **extremely time-consuming** and requires domain expertise, limiting the frequency of transparency score updates.

***Can LLMs be used to help extract and categorize methodological details, for more timely and consistent transparency scoring?***




Mark Blumenthal found that election pollsters that disclosed more about their methodology and fieldwork in 2022 had less error

# Ground Truth

Dataset

We use a **2024 transparency scoring dataset compiled by FiveThirtyEight and Mark Blumenthal**, which includes detailed assessments of how polling organizations adhere to AAPOR transparency guidelines.

It was manually compiled by former 538 researchers Mary Radcliffe and Cooper Burton, and serves as the benchmark for evaluating the LLMs' ability to extract comparable methodological details.

abc NEWS Log In

1. Did the pollster publish the exact trial-heat question wording used in this poll?
2. Did the pollster publish the exact question wording and response options for every question mentioned in the poll release?
3. Did the pollster release both weighted and unweighted sample sizes for any demographic groups, or acknowledge the existence of a design effect in their data?
4. Did the pollster publish crosstabs for every subgroup mentioned in the poll release?
5. Did the pollster disclose the sponsor of the poll (if there was a sponsor)?
6. Did the poll specify how the sample was selected (e.g., via a probability-based or non-probability method)? If the sample was probability, was the sampling frame disclosed? If non-probability, did the pollster disclose what marketplace or online panels were used to recruit responses or its model for respondent selection?
7. Did the pollster list at least three of the variables the poll is weighted on?
8. Did the pollster disclose the source of its weighting targets (e.g., "the 2022 American Community Survey")?
9. Did the poll report a margin of error or sample size for a "critical mass" of subgroups? We do not mandate this be a complete count, but if it looks like groups are intentionally missing (e.g., they are referenced in the press release but are missing in the crosstab documents), we withhold the point.
10. Did the poll methodology or release include a general statement acknowledging a source of non-sampling error, such as question wording bias, coverage error, etc., in addition to the normal margin of sampling error inherent to surveying?

# Known Challenges

### With LLMs

Large language models often hallucinate information or misinterpret ambiguous phrasing

Output can vary depending on model settings (e.g., temperature), impacting reliability.

### Measuring Transparency Scores

Data is present in multiple format (e.g., PDFs, tables, images)

Coding methodological elements can involve subjective judgment (subsample MoE, disclosure of sponsor)

### Limitations of the Study

Researchers still need locate and validate the relevant links for each poll, and convert the link into a format that the LLM can read

# What we are collecting

**Measuring the LLMs ability to collect and structure the following information about polls to evaluate methodological transparency across pollsters.**

For PollFinder De-Duplication

1. Name of the polling firm or organization.
2. Polling method (e.g., telephone, online, IVR, mixed mode).
3. Names of sponsors, if disclosed.
4. Start and end dates of polling.
5. Geographic scope (e.g., national, state-level, local).

For Transparency Score

1. Does the poll disclose who sponsored it? *(true/false)*
2. Is the trial heat question asked? *(true/false)*
3. Are full question wordings provided? *(true/false)*
4. Is the source of the sample described? *(true/false)*
5. Are variables used for weighting disclosed? *(true/false)*
6. Are target values for weighting disclosed? *(true/false)*
7. Are unweighted and weighted sample size listed. *(true/false)*
8. Is the design effect mentioned? *(true/false)*
9. Are crosstabs available? *(true/false)*
10. Reports subgroup-level size or margin of error. *(true/false)*

# Experiment Design

**Sample Selection**

To reduce the impact of human labeling errors filtered for **unique pollster-sponsor pairs**.

We also excluded polls with inactive or broken links, ensuring all inputs were accessible.
(804 polls → 156 polls)

**Experiment**

For each poll, we provided the LLM (GPT-4o model) with all documents and links listed in the FiveThirtyEight dataset in a PDF or xlsx format

The LLM was prompted to output:

- The value (in 538's dataset's original format)
- The supporting sentence from the source document

**Review**

We manually reviewed a sample of the polls for accuracy and iteratively re-prompted the LLM to improve outputs

| *LLM Answer* | *source sentence* | *538 Answer* | *Dhrumil Answer* | *Correct?* |
|---|---|---|---|---|
| FALSE | "The sample size for | FALSE | FALSE | Correct (W… |
| FALSE | Only "582 likely voters" | FALSE | FALSE | Correct (W… |
| TRUE | "We surveyed 1,086 elig "design_effect": "The | FALSE | FALSE | Wrong |
| FALSE | Only one figure is menti | FALSE | FALSE | Correct (Ri… |
| TRUE | "The latest Reuters/Ip "The data were weigh | FALSE | FALSE | Wrong |
| FALSE | N=700 respondents" — | FALSE | FALSE | Correct (W… |
| FALSE | Only "N = 419" is given | FALSE | FALSE | Correct (W… |
| TRUE | "A total of 739 surveys "Q1. In what county | TRUE | TRUE | Correct (Ri… |
| FALSE | "Total Number of Respo "The sample was we | FALSE | FALSE | Correct (W… |
| TRUE | "This report includes 76 "The online sampling | FALSE | FALSE | Wrong |
| FALSE | Only one N reported ("1 | FALSE | FALSE | Correct (W… |

The edit process involved a manual review of each response by the LLM

# Prompt and Response Structure

**Example Descriptions:**

```
"disclose_sponsor": {
  "type": "boolean",
  "description": "Indicates whether the poll discloses who sponsored it."
},
"sponsor_names": {
  "type": "string",
  "description": "Names of the sponsors."
},
"start_date": {
  "type": "date",
  "description": "Date the poll was conducted (start date)."
},
```

**Example Response**


```
{
  "pollster_name": "Research Co.",
  "mode": "online",
  "disclose_sponsor": true,
  "sponsor_names": "Research Co.",
  "start_date": "2024-11-02",
  "end_date": "2024-11-03",
  "state": "Pennsylvania",
  "trial_heat_question_wording": true,
  "all_questions_wording": true,
  "source_of_sample_data": true,
  "weighting_variables": true,
  "weighting_targets": true,
  "weighted_and_unweighted_ns": true,
  "design_effect": 1.0,
  "publish_crosstabs": true,
  "subsample_size_or_moe": true,
  "source_sentences": {
    "pollster_name": "Polls conducted by Research Co. in the United States - November 4, 2024",
    "mode": "Methodology: Results are based on online surveys conducted on November 2 and November 3, 202
    "disclose_sponsor": "Polls conducted by Research Co. in the United States - November 4, 2024",
    "sponsor_names": "Polls conducted by Research Co. in the United States - November 4, 2024",
    "start_date": "Methodology: Results are based on online surveys conducted on November 2 and November
    "end_date": "Methodology: Results are based on online surveys conducted on November 2 and November 3,
    "state": "Base: 450 voters in Pennsylvania",
    "trial_heat_question_wording": "In the presidential election scheduled for Nov. 5, which one of the
    "all_questions_wording": "Which one of the following candidates would you be most likely to support
    "source_of_sample_data": "The data has been statistically weighted according to U.S. census figures
    "weighting_variables": "The data has been statistically weighted according to U.S. census figures fo
    "weighting_targets": "The data has been statistically weighted according to U.S. census figures for
    "weighted_and_unweighted_ns": [
      "Base: 450 voters in Pennsylvania",
      "Base: 433 decided voters in Pennsylvania"
    ],
    "design_effect": "The margin of error... is +/- 4.6 percentage points for each state.",
    "publish_crosstabs": "Breakdowns provided by gender, age, party ID, and 2020 vote for all candidates
    "subsample_size_or_moe": "Breakdowns of results for subgroups (e.g., gender, age, political ID) with
  }
}
```

# Results for subsample

*156 Items, LLM 🤖 vs manual evaluation by Mary and Cooper 🤓*

| | 🤓 Positive | 🤓 Negative |
|---|---|---|
| 🤖 Neg | False positive | True negative |
| 🤖 Pos | True positive | False negative |

**Pollster Name**
accuracy: 154/156 = **98.7%**

**Polling method**
accuracy: 136/156 = **87.1%**

**Names of sponsors***
accuracy: 67/80 = **83.8%**

**Start and end dates**
accuracy: 149/156 = **95.5%**

**Geographic scope**
accuracy: 155/156 = **99.3%**

| | | |
|---|---|---|
| **Sponsors Disclosed**<br>accuracy: 137/156 = **87.8%** | 1 | 1 |
| | 136 | 18 |
| **Trial Heat Question Present**<br>accuracy: 121/156 = **77.6%** | 5 | 29 |
| | 92 | 30 |
| **Wording for All Questions**<br>accuracy: 120 / 156 = **76.9%** | 2 | 32 |
| | 88 | 34 |
| **Source of Sample Data**<br>accuracy: 114/156 = **73.1%** | 36 | 28 |
| | 86 | 6 |
| **Weighting Variables**<br>accuracy: 123/156 = **78.8%** | 21 | 28 |
| | 95 | 12 |

| | | |
|---|---|---|
| **Weighting Targets**<br>accuracy: 102/156 = **65.4%** | 45 | 34 |
| | 68 | 9 |
| **Weighted and Unweighted (ns)**<br>accuracy: 25/156 = 16.0%<br>*After prompt engineering:* 22/30 = **73.3%** | 129 | 9 |
| | 16 | 2 |
| **Design Effect**<br>accuracy: 136/156 = **87.1%** | 2 | 118 |
| | 18 | 18 |
| **Crosstabs Published ****<br>accuracy: 127/156 = **81.4%** | 10 | 30 |
| | 97 | 19 |
| **Subsample Size or MOE**<br>accuracy: 79/156 = 50.6%<br>*After prompt engineering:* 23/30 = **76.7%** | 77 | 0 |
| | 79 | 0 |

* *Only evaluated if the manual evaluation identified a sponsor*

** *When crosstab documents existed, they were uploaded to the LLM. This does not evaluate if the LLM can crawl the web to find crosstab documents.*

# Iterating on the prompts

**Issues**

The LLM performed well when information was explicitly stated in the text. However, it struggled to identify unweighted and weighted sample size and disambiguate subsample size and MOE and from the overall sample size and MOE.

**Prompting for More Accuracy**

After each attempt with the prompt, we manually reviewed a sample of the entries, noting patterns and specific issues related to the accuracy of the returned data and iterated accordingly

**Prompt Attempt 1**

Does this poll report the size and/or margin of error (MOE) of any subgroups?

**Prompt Attempt 3:**

*Does this poll report the size and/or margin of error (MOE) for any demographic or political subgroups? The information may be embedded in tables or charts rather than stated outright.*

- If subsample sizes are included (e.g., as unweighted or weighted N values), list them by subgroup.
- If margins of error are included for any subsamples, report them explicitly.
- If either is missing, state that clearly.

**Important:** Do not confuse the overall sample size or MOE (e.g., "N=1,000, MOE ±3%") with the subgroup-specific figures.

**Prompt Attempt 5:**

TRUE if Size of subgroups OR Margin of Error. Does this poll report the size and/or margin of error (MOE) for any demographic or political subgroups (e.g., gender, age, race, education, partisanship, etc.)?

DO NOT confuse the overall sample size or MOE (e.g., "total sample...,likely voters N=1,000, MOE ±3%") with the subgroup-specific figures. Focus only on crosstab-level reporting.

# Subsample Size or MOE

AFTER PROMPT ENGINEERING

| | 🤓 Positive | 🤓 Negative |
|---|---|---|
| 🤖 Neg | False positive | True negative |
| 🤖 Pos | True positive | False negative |

*Manual Evaluation by Dhrumil & Mark*

**First Prompt**

Does this poll report the size and/or margin of error (MOE) of any subgroups?

| | |
|---|---|
| 0 | 0 |
| 15 | 15 |

| Right for Right "Reason" | Right for Wrong "Reason" | Wrong |
|---|---|---|
| 0 | 15 | 15 |

Accuracy: 15/30 = **50%**

**Last Prompt**

TRUE if Size of subgroups OR Margin of Error. Does this poll report the size and/or margin of error (MOE) for any demographic or political subgroups (e.g., gender, age, race, education, partisanship, etc.)?

DO NOT confuse the overall sample size or MOE (e.g., "total sample...,likely voters N=1,000, MOE ±3%") with the subgroup-specific figures. Focus only on crosstab-level reporting.

| | |
|---|---|
| 6 | 9 |
| 14 | 1 |

| Right for Right "Reason" | Right for Wrong "Reason" | Wrong |
|---|---|---|
| 15 | 8 | 7 |

Accuracy: 23/30 = **76.7%**

# **Example:** Subsample Size or MOE

| | Right for Right "Reason" | Right for Wrong "Reason" | Wrong |
|---|---|---|---|
| 🤖 **LLM's Response** | The 'Results' sheet in the file provides explicit unweighted sample sizes (Ns) for many subgroups. For example:<br><br>Party ID: Democrat = 213, Republican = 112, Independent/Other = 245<br><br>Gender: Men = 263, Women = 311<br><br>Age groups: 18–29 = 63, 30–44 = 168, etc.<br><br>These appear in row 2 under "Unweighted count" and align with the detailed subgroup labels above each column. | The poll includes explicit numerical or percentage-based breakdowns for multiple subgroups, such as:<br><br>"Harris leads Trump among women by eight percentage points, 52% to 44%... Trump holds a much larger lead among men... 56% to 38%"<br><br>"Trump... leads Harris by ten percentage points, 54% to 44%" among seniors (65+) "Among African American women... 88% to 5%" "Among African American men... 74% to 22%"<br><br>— These indicate that responses are being quantified by subgroup... | Crosstabs like: "Men: Trump 53%, Harris 41%" and "College Grads: Harris 55%, Trump 41%" |
| 🤓 **Our Evaluation** | • LLM is reading the correct part of the spreadsheet and finding the subsample size | • Sample n's are provided but the LLM only references percentages | • LLM mentions percentages and marks subsample_size_or_moe as TRUE, but sample n's are not provided |

# Weighted and Unweighted Sample Size

AFTER PROMPT ENGINEERING

| | 🤓 Positive | 🤓 Negative |
|---|---|---|
| 🤖 Neg | False positive | True negative |
| 🤖 Pos | True positive | False negative |

*Manual Evaluation by Dhrumil & Mark*

**First Prompt**

"Does this poll report the weighted and unweighted sample sizes?"

| | |
|---|---|
| 1 | 6 |
| 2 | 21 |

| Right for Right "Reason" | Right for Wrong "Reason" | Wrong |
|---|---|---|
| 4 | 4 | 22 |

Accuracy: 8/30 = 26.7%

**Last Prompt**

"Mark as true only if both the unweighted and weighted sample sizes are explicitly listed numerically.

Descriptive explanations of weighting procedures or inclusion of a design effect are not sufficient on their own.

If only an unweighted N is provided (e.g., "N=1,000") and there is no explicit numerical mention of the weighted N, this should be marked false."

| | |
|---|---|
| 7 | 20 |
| 2 | 1 |

| Right for Right "Reason" | Right for Wrong "Reason" | Wrong |
|---|---|---|
| 4 | 18 | 8 |

Accuracy: 22/30 = 73.3%

# **Example:** Weighted and Unweighted Sample Size

| | Right for Right "Reason" | Right for Wrong "Reason" | Wrong |
|---|---|---|---|
| 🤖 **LLM's Response** | "A total of 739 surveys were completed from a random selection of adults in Wyoming.",<br><br>"Q1. In what county in Wyoming do you live?... Unwtd Percent... Weighted Percent",<br><br>"Weighting... The variables used in the weighting procedure were... [with complete tables showing both unweighted and weighted counts throughout the report]" | Only unweighted N (=1005) is provided; no weighted N or explanation of how weighting affected estimates — mark as false | "This report includes 760 responses...",<br>"The online sampling method collected 526 responses, and 234 responses were collected via phone.",<br>"Results are weighted based on Census data." |
| 🤓 **Our Evaluation** | • Correctly looks at crosstab tables | • Actually 1005 is the weighted N<br>• This transparency metrics has to do with crosstab samples, but LLM is looking at topline sample | • Seems to be marked true based on the presence of the word "weighted" but no weighted/unweighted subsample # provided. |

# **Example:** Is the sponsor disclosed?

| | Did Pollster Disclose Sponsor? |
|---|---|
| 🤖 **LLM's Response** | "The document does not specify who sponsored the poll, only that it was conducted by Axis Research." **FALSE** |
| 🤓 **Human Response** | This is an internal NRCC poll **TRUE** |

## Poll Memo

AXIS RESEARCH

### Methodology

The results presented above are taken from three districtwide polls fielded with likely 2024 voters. The most recent poll fielded from October 17-20 with 411 likely general election voters. The poll was fielded through a mixed methodology: 71% cell phone interviews with a live interviewer | 17% text-to-web interviews | 13% landline interviews with a live interviewer.

The margin of error for each poll is ±4.8 percentage points.

## Politico Article

**FIRST IN PLAYBOOK —** A new internal NRCC poll finds **AUSTIN THERIAULT** narrowly ahead of Rep. **JARED GOLDEN** (D-Maine) in a crucial swing district, 47 percent to 45 percent. They also have Trump leading in the district, which awards its own electoral vote, by 9 points. Both numbers have moved toward the GOP since their last survey, as Golden's lead among independents has vanished. ***The polling memo***

# Conclusion

**Good enough for a "first pass"**

This can be helpful for an editor with lots of domain knowledge Every single cell still needs to be carefully manually checked.

**Limitations**

An additional "web crawling" agent is needed to find the link containing methodology details ie. crosstabs, toplines

**Can be used to find new polls**

The language model is able to identify the name of the pollster, start and end dates, geographic scope with decent level accuracy.

These attributes can be used to determine if a poll is unique.

High failure tolerance as long as there is a human in the loop.

## Poll Detector

### Detecting poll mentions using LLMs 🔬

This app processes Google Alerts and Search results to identify political polling data, using large language models to extract structured poll metadata.

Use the filters below to refine your search based on date, poll mentions, poll count, or specific collections like Trump or Musk approval.

**Last Updated:** May 15, 2025 at 1:42 PM

**Date:** 05/12/2025 12:00 AM

**Only Polls Mentioned:**

**Max # Polls:** 1

**Collection:** poll AND (approve AND Trump)

| Added On (Local T... | Collections | Pollster | Article URL | P.. | # ... |
|---|---|---|---|---|---|
| 5/12/2025, 7:38:59 PM | poll AND (approve AND Trump) | UC Berkeley's Institute of Governmental Studies | capradio.org... | ✅ | 1 |
| 5/12/2025, 6:24:04 PM | poll AND (approve AND Trump) | Fox News | wfin.com... | ✅ | 1 |
| 5/12/2025, 11:20:01 AM | poll AND (approve AND Trump) | GrayHouse | silive.com... | ✅ | 1 |
| 5/12/2025, 10:19:01 AM | poll AND (approve AND Trump) | AP-NORC | debatepolitics.com... | ✅ | 1 |

*Screenshot from the https://www.pollfinder.ai/ prototype app, which crawls the web to find polls hourly. LLM-identified metadata is used to de-duplicate links referring to the same poll.*

# Thank You!

Want updates on this work?
Sign up at https://www.pollfinder.ai/

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